What Makes a Good ML Loss Function?

Optimizable

- Be differentiable (preferably smooth)
- Provide informative gradients for optimization
- Allow for principled regularization (e.g., via convexity or smoothness)

Practical & Scalable

- Scale well computationally (especially in deep learning)
- Work with mini-batching and SGD
- Integrate easily into modern ML pipelines

Aligned with the Learning Task

- Align well with the task's performance metric
- Be robust to noise, outliers, misspecification
- Reflect the structure of the underlying data domain

EOT Derivatives via Autodiff